Do Now · retrieval

Scope & parameter passing in C#

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the difference between a C# function and a procedure-style method?

Q2

Which C# loop always executes its body at least once?

Q3

What does == do in a C# condition?

Q4

In Triple(7) , identify the argument. In Triple(int number) , identify the parameter.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Scope controls where a variable can be used — A local variable is declared inside a method or block and can only be used within that scope. — For this course, we use a class-level static field to demonstrate the OCR idea of a global variable that can be accessed by multiple methods.

Core idea 2

Parameters are local to the method — A parameter receives data when a method is called and behaves as a local variable inside that method. — CODE:

```
public static int Triple(int number)
{
return number * 3;
}
```

Core idea 3

By value = the method receives a copy — CODE:

```
public static void Change(int number)
{
number = 100;
}
```

— Changing number changes only the method's local copy. The original variable in Main() is unchanged.

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

ref gives access to the original variable — CODE:

```
public static void Change(ref int number)
{
number = 100;
}

// call
Change(ref value);
```

— Write ref in both places: in the method parameter and in the method call.

Vocabulary · use the correct technical terms

Scope — The part of a program in which a variable can be accessed.

Local variable — A variable visible only within the method or block where it is declared.

Global variable — A variable that can be accessed from multiple parts of a program.

Parameter — The named input in a method definition.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Why doesn't the original value change?

Follow the worked method

1. Start in Main — CODE:
int score = 10; — The variable score belongs to Main() and currently stores 10.

2. Only number becomes 100 — CODE:
ChangeByValue(score);

inside method:
number = 100

back in Main:
score = 10 — When the method ends, its local parameter is no longer needed.

3. Reference passing changes the original — CODE:
ChangeByReference(ref score);

```
// method
public static void ChangeByReference(ref int number)
{
number = 100;
}
```

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Classify each variable

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Local variables in two methods
Create two procedure-style methods: First() and Second() .
Inside each method declare a local variable called result , but give it a different value. Output each value.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Swap two values
Create Swap(ref int first, ref int second) .
Use one local temporary variable so that the two original values exchange places.
In Main() , start with a = 10 and b = 25 , call the method and output both values.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Passing an integer parameter by value means changing the parameter does not change the original integer variable.

Use key vocabulary from memory

1. Explain Scope in your own words.

2. Explain Local variable in your own words.

3. Explain Global variable in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.